

Rental Platform

1. Introduction

Purpose

This document explains the requirements of the Rental Platform, a website where people can give camera equipment on rent and other people can book it for the days they need.

Scope

The Rental Platform lets an owner add camera gear like cameras, lenses, lights and tripods with a price per day. A renter can search the gear, choose the dates and send a booking request. The website shows an availability calendar, so two people cannot book the same item on the same dates. It also uses Google Gemini AI to write simple rental terms for the owner.

Online payment, chat, reviews and delivery are not included in this project.

Definitions

- Owner: a user who gives equipment on rent.
- Renter: a user who books equipment. One account can be both owner and renter.
- Listing: one item that is available for rent.
- Booking: a request to rent a listing from a start date to an end date.
- Availability calendar: a calendar that shows which dates of an item are free or booked.
- Gemini AI: Google's AI, used here to write the rental terms.
- SQLite: a simple database that stores all data in a single file.

2. Overall Description

Product Perspective

The Rental Platform is a web application. The frontend is made with HTML, CSS and JavaScript, the backend is made with Node.js and Express, and the data is stored in an SQLite database. The Gemini AI is called from the backend so that the API key stays safe.

Product Functions

- Register, login and edit profile
- Add, edit, delete, search and browse equipment
- Check availability and book dates without overlap
- Accept, reject or cancel bookings
- Generate rental terms using AI

User Classes

- Owner: adds equipment and handles booking requests.
- Renter: searches equipment and books it.
- Admin (optional): removes wrong or inappropriate listings.

Operating Environment

The website works in common browsers like Chrome, Edge and Firefox. The server needs Node.js. The SQLite database is a single file, so no separate database server is needed.

Constraints

- The project uses HTML, CSS, JavaScript, Node.js, Express, SQLite and the Gemini API.
- Prices are in rupees and are charged per day.
- Payment between owner and renter is done outside the website.
- The project has to be completed in a short time.

Assumptions

- Users have an internet connection.
- A Gemini API key is available.
- Both the start date and the end date are counted as rental days.

3. Specific Requirements

The requirements are divided into five modules. Points marked (optional) will be done only if time is left.

Authentication and Profile

- A user can register with name, email, phone number and password.
- A user can login with email and password.
- Passwords are stored in a safe (hashed) form.
- Some pages can be opened only after login.
- A user can logout.
- A user can edit his profile (optional).

Listings and Search

- An owner can add a listing with title, description, category, price per day, city and images.
- An owner can edit or delete only his own listings.
- A listing that has upcoming bookings cannot be deleted.
- An owner can make a listing active or inactive.
- All active listings are shown on the browse page.
- Users can search by keyword and filter by category, city and price.
- The detail page of a listing shows item details, rental terms, calendar and booking form.
- An owner can see all his listings on the My Listings page.

Availability Calendar

- Each listing has a calendar in which booked dates are disabled.
- Past dates cannot be selected.

- The server rejects a booking if its dates overlap with another booking.
- An owner can block some dates, for example when the item is under repair (optional).
- When a booking is rejected or cancelled, its dates become free again.

Booking Management

- The renter selects the start date and end date and sees the total price.
- After submitting, the booking is created with status Pending.
- The owner can accept or reject the booking.
- The renter can cancel a booking before the start date.
- The booking status can be Pending, Confirmed, Rejected, Cancelled or Completed.
- The renter can see My Bookings and the owner can see Booking Requests.
- A user cannot book his own listing.
- The renter must accept the rental terms before booking.

AI Rental Terms

- The owner can click a Generate Rental Terms button while adding a listing.
- The AI writes simple terms about use of the item, return time, damage, late fee and cancellation.
- The owner can edit the terms and then save them.
- The Gemini API key is kept only on the server.
- If the AI does not work, a default terms template is shown.
- A note is shown that the terms are AI generated and not legal advice.

4. External Interface Requirements

User Interface

The website has these pages: Register, Login, Browse, Listing Detail, Add or Edit Listing, My Listings, My Bookings, Booking Requests and Profile. It should work on both mobile and computer screens.

Software Interface

- SQLite database
- Google Gemini API

5. Non-Functional Requirements

- Performance: pages should load quickly, in about 3 seconds.
- Security: passwords are hashed, login is checked with a token and secret keys are kept in the .env file.
- Usability: the website should be simple and easy to use on mobile.
- Reliability: the same item must never be booked twice for the same dates, and if the AI fails, booking should still work.

6. Database Design

- User: name, email, phone, city, password.
- Listing: owner, title, description, category, price per day, city, images, rental terms, active status.
- Booking: listing, renter, owner, start date, end date, total price, status.

7. Future Scope

- Online payment
- Chat between owner and renter
- Reviews and ratings
- Email and SMS notifications
- More categories like costumes and sports equipment